



PHARMACEUTICAL DOSAGE FORMS II- PPC 202

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Lecture (6)

Semisolid Dosage Forms

Pastes & Poultices



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Intended Learning Outcomes “ILO’s”

After finishing “Lecture 6” the student should be able to:

- Differentiate between **pastes** & **poultices**
- Identify **miscellaneous types** of dermal applications
- Define **ideal SSDF**

Lecture content:

After finishing “Lecture 6” the student should be able to:

- **Types** of Paste & Poultice **bases**
- **Miscellaneous** types of dermal applications

Types of topical semisolid preparations

Semisolids

Ointments

Creams

Gels

Pastes

Poultices



Pastes

- **Pastes** are semisolid preparations into which a high percentage (**at least 25%**) of insoluble solid has been added, so it is usually **stiffer than ointments**
- **Pastes** make particularly good **protective barrier** when placed on the **skin**, the **solid** they contain can **absorb exudates**



Pastes



Types of bases:

1. Hydrocarbon base (Fatty base):

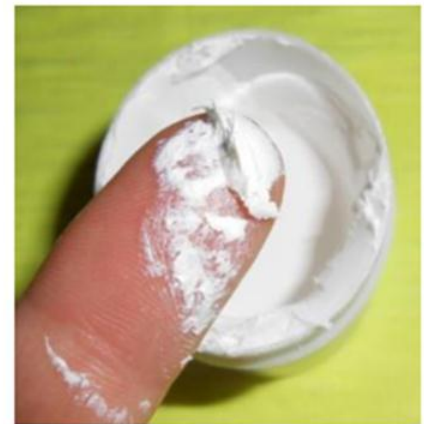
- E. g. Lassar's paste:

Active ingredients:

- Zinc Oxide (24%), and Salicylic acid (2%)

Excipients: Starch, soft paraffin, and liquid paraffin

- It is used to treat and prevent **diaper rash** and other minor **skin irritations** (e.g., burns, cuts, ...). It works by forming a protective barrier on the skin to protect it from irritants/moisture.



Pastes



Types of bases:

2. Water miscible base:

E. g. Magnesium Sulphate paste:

- Magnesium sulphate 48% w/w mixed with 0.5% **phenol in glycerol**.
- It is used for treating **boils**, withdrawing exudates. Known in the UK as "**drawing paste**"



Pastes

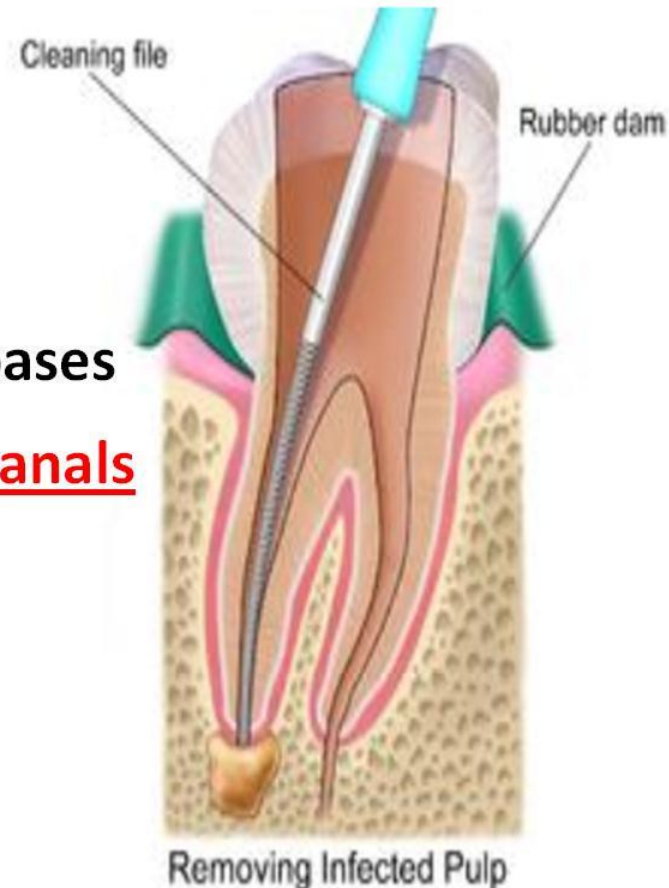


Types of bases:

3. Water soluble base:

E. g. Dental Pastes:

- Containing **neomycin sulphate** and **PEG bases**
- It is used for **Sterilizing infected root canals**



Types of topical semisolid preparations

Semisolids

Ointments

Creams

Gels

Pastes

Poultices



Poultices (Cataplasms)

- It is **soft, viscous, pasty** preparation for **external** use. They are applied to skin while they **are hot** and kept in place with **a cloth**.
- Poultice must retain heat for a considerable time **because:** “they are intended to supply warmth to inflamed parts of body”.
- They are used **for boils, carbuncles and inflamed areas** for e.g. Kaolin Poultice
- **Turmeric** **كرم** and **ginger** can be mixed to make a **paste**, or boiled together along side a clean washcloth in salted water, and applied to the **boil** for 5 to 10 minutes daily.



Miscellaneous types of dermal applications

Plasters

Plasters are **solid or semisolid masses adhere to the skin** when spread upon **cotton felt line** or muslin انسج as a backing material



Salonpas

backache, sore shoulders, & muscular aches



Medicated Plaster

Salicylic Acid Plaster



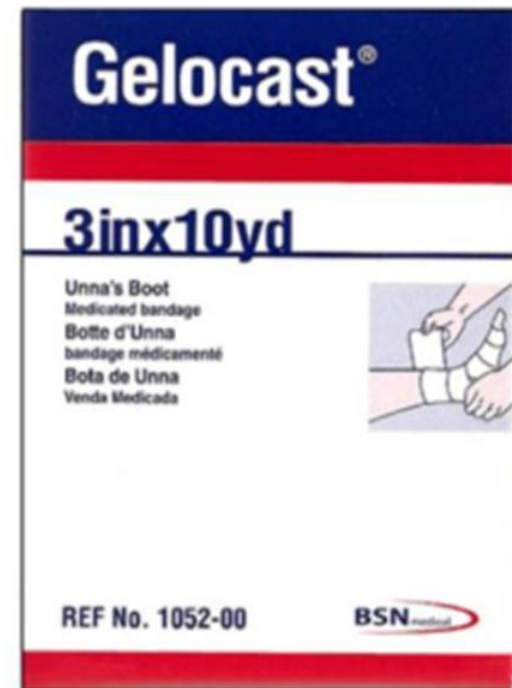
Glycerogelatins

- Plastic masses containing: **gelatin** (15%), **glycerin** (40%), **water** (10%) & an added **medicinal substance** (10%)

- **Example:**

- 1. Zinc Gelatin Boot**

- A plastic mass containing 10% **zinc oxide** in a glycerogelatin base
- **For the treatment of varicose ulcers**



Some terms commonly used to describe the properties of semisolid preparations

1. Occlusive

- **Prevent the water loss** (evaporation) from the skin leading to skin hydration, loose skin structure, increase skin porosity leading to **increase in drug permeation**.
- **Occlusive products** can be used **as moisturizers** (increase skin hydration) or to **increase drug absorption** (e.g. **hydrocortisone ointment**)
- **Occlusive materials**: fats, paraffin, oils (most fatty bases)

2. Greasy & Emollient

Greasy

- ▶ Materials providing **oily texture to the skin**
- ▶ e.g.: fatty bases

Emollient

- ▶ Increase skin **elasticity & softening**
- ▶ e.g.: vegetable oils, lanolin

3. Moisturizer & Spreadability

Moisturizer

- ▶ Increase the moisture (water) content of the skin
- ▶ e.g.: **occlusive substances**, some chemical substances as **lactic acid** and **urea**, as well as **humectants** (**Glycerol**)

Spreadability

- ▶ Spreading the product on the skin
- ▶ Easily spread materials **good spreading**
- ▶ Difficult spread material **bad spreading**

5. Drug Release & Permeation Penetration

Drug Release

- ▶ Availability of the drug to the skin
- ▶ **drug diffuses through the **base** and become in contact with the skin surface**

Permeation Penetration

- ▶ Ability of the drug to **penetrate through the skin**
- ▶ **passing the stratum cornea, reaching the epidermis, dermis, subcutaneous or systemic circulation**

Important notes

You should be aware of ??????????

- A **topical product** should never be applied over open wounds or broken skin unless it is labeled as a sterile product.
- Besides the risk of **infection**, such an application is often associated with problems of significant **systemic drug absorption**

Midterm Exam

Lectures Topics Included: week 1 till week 5

Skin Structure

Ointment

Creams & Gels

Question Types:

MCQ

True or False

*Thank
you*



END OF TOPICAL SEMISOLID DOSAGE FORMS